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Estimating Discount Functions with Consumption Choices over the Lifecycle

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1 Big picture

- **Question.** Can household lifecycle balance sheets identify short-run and long-run discounting, especially when households both revolve high-interest credit card debt and accumulate substantial illiquid wealth?
- **Core empirical tension.** Many households appear impatient when we look at credit card borrowing, but patient when we look at retirement, housing, and other illiquid wealth.
- **Main method.** The paper estimates a structural lifecycle model by matching sixteen balance-sheet moments: four age groups crossed with four moment blocks.
- **Main result.** The benchmark present-biased specification estimates $\hat{\beta} = 0.5305$, $\hat{\delta} = 0.9891$, and relative risk aversion $\hat{\rho} = 1.9355$.
- **Economic meaning.** The estimates imply a very high short-run discount rate and a low long-run discount rate. This combination rationalizes borrowing on expensive credit cards while accumulating illiquid assets.
- **Exponential benchmark.** Imposing $\beta = 1$ produces $\hat{\delta} = 0.9601$ and $\hat{\rho} = 1.4663$, but the model badly underpredicts credit card borrowing.
- **Identification insight.** Credit card moments want a low β and high short-run impatience; wealth moments want a high δ and patience over longer horizons.
- **Boundary analysis.** The paper introduces a graphical boundary method that shows which regions of (β, δ, ρ) fit credit-card moments and wealth moments separately.
- **Policy relevance.** The estimates matter for models of retirement saving, consumer credit, liquidity regulation, credit-card borrowing, and welfare analysis of high-interest loans.

2 Incremental contribution of the paper

- **Moves beyond small-stakes experiments.** Earlier work often estimates time preferences from laboratory or field choices over small or artificial stakes. This paper estimates discounting from lifecycle choices with large financial consequences.
- **Uses the whole balance sheet.** The paper does not infer impatience from credit card debt alone or patience from retirement wealth alone. It jointly targets debt and wealth moments, conditional on age and debt status.
- **Connects present bias to the “debt-and-wealth” puzzle.** The paper gives a structural interpretation of why households can simultaneously hold costly revolving credit card debt and substantial illiquid wealth.
- **Brings present-biased preferences into a rich quantitative lifecycle model.** The model includes stochastic income, liquid and illiquid assets, credit cards, dependents, Social Security, mortality, and bequests.
- **Improves on calibration by using two-stage MSM.** Preference parameters are estimated by minimizing a formal moment distance, while uncertainty in first-stage parameters and simulation error is propagated into standard errors.
- **Provides a new identification diagnostic.** The boundary analysis visualizes how subsets of moments identify different regions of the parameter space, making the structural estimation more transparent.
- **Clarifies when exponential discounting can work.** Exponential discounting can match borrowing and wealth only when the credit-card borrowing rate is much lower or the illiquid return is much higher; under the benchmark market rates, present bias is needed.
- **Shows robustness beyond a single interpretation.** Heterogeneity, sophistication, and educational subgroup exercises suggest the finding $\beta < 1$ does not rely only on the benchmark naivete assumption.

3 Data and measurement

3.1 Second-stage moments from household balance sheets

The paper uses the Survey of Consumer Finances to construct four moment blocks for households whose heads have a high school degree and no college degree:

- *%Visa*: the share of households revolving high-interest credit card debt;
- *meanVisa*: average carried credit card debt, including zeroes;
- *wealth | debt*: average net wealth among households with revolving credit card debt;
- *wealth | no debt*: average net wealth among households without revolving credit card debt.

Each block is measured in four age intervals: 21-30, 31-40, 41-50, and 51-60. These generate the sixteen moments used to identify the second-stage preference parameters.

Interpretation: The age dimension is essential. The model must explain not only average borrowing and wealth, but how borrowing and wealth evolve over the lifecycle.

3.2 First-stage parameters and model inputs

The first-stage parameter vector χ includes income-process parameters, household composition profiles, credit limits, and calibrated return parameters. Table 2 shows the first-stage estimates.

- Income follows a life-cycle profile with age, household-size terms, persistent income shocks, and transitory shocks.
- Household size varies with children and dependent adults.
- Liquid assets earn a lower return when positive and become credit card debt when negative.
- Illiquid assets generate a consumption flow and are costly to liquidate.

Interpretation: These first-stage objects discipline the budget constraints before preference parameters are estimated. The model is not allowed to fit borrowing by arbitrarily changing income risk or credit limits in the second stage.

3.3 The meaning of the credit-card moments

The credit-card moments summarize both extensive and intensive borrowing:

- *%Visa* is the probability of revolving debt;
- *meanVisa* is the average amount of revolving debt relative to income.

In the data, more than half of households in several working-age groups carry revolving credit-card debt.

Interpretation: High revolving debt at an 11 percent real borrowing rate is difficult to rationalize with a patient exponential model, because borrowing is expensive relative to the return on illiquid wealth.

3.4 The meaning of the wealth moments

The wealth moments are scaled by income and separated by debt status. A key empirical fact is that even households with credit card debt have substantial net wealth, especially at older ages.

Interpretation: The coexistence of credit-card debt and illiquid wealth is the empirical core of the paper. It pushes the model toward preferences that are impatient today but patient about future accumulation.

4 Lifecycle model and estimating equations

4.1 Liquid and illiquid asset dynamics

Let X_t be liquid wealth and Z_t illiquid wealth. Net investments are I_t^X and I_t^Z . The asset laws of motion are:

$$X_{t+1} = R^X X_t + I_t^X, \quad (1)$$

$$Z_{t+1} = R^Z Z_t + I_t^Z. \quad (2)$$

The liquid return depends on whether the household is borrowing:

$$R^X = \begin{cases} R^{CC}, & X_t + I_t^X < 0, \\ R, & X_t + I_t^X \geq 0. \end{cases} \quad (3)$$

Consumption is pinned down by income, investment, and liquidation of illiquid assets:

$$C_t = Y_t - I_t^X - I_t^Z + \kappa_t \min\{I_t^Z, 0\}. \quad (4)$$

Interpretation: Negative liquid wealth is credit-card borrowing and carries the high credit-card rate R^{CC} . The illiquid asset earns a premium but is costly or slow to access.

4.2 Utility and household size

The model uses CRRA flow utility with effective household size n_t :

$$u(C_t, Z_t, n_t) = n_t \frac{\left(\frac{C_t + \gamma Z_t}{n_t}\right)^{1-\rho} - 1}{1-\rho}. \quad (5)$$

The term γZ_t captures the consumption flow from illiquid assets such as housing or retirement wealth.

Interpretation: The household values liquid consumption and the service flow from illiquid assets. This is why a household can rationally accumulate illiquid wealth even while it is sometimes liquidity constrained.

4.3 Present-biased discount function

The discount function is

$$1, \quad \beta\delta, \quad \beta\delta^2, \quad \beta\delta^3, \dots \quad (6)$$

When $\beta = 1$, the model reduces to exponential discounting. When $\beta < 1$, the household is more impatient over the immediate horizon than over all later horizons.

Interpretation: β identifies short-run impatience, while δ identifies longer-run patience. The paper's key result is that both are needed to fit the balance sheet moments.

4.4 Naive present bias

Under the benchmark naivete assumption, self t chooses current investment to maximize

$$u(C_t, Z_t, n_t) + \beta\delta E_t V_{t,t+1}^E(\Lambda_{t+1}). \quad (7)$$

Self t incorrectly believes future selves will behave exponentially, effectively setting $\beta = 1$ for future selves.

Interpretation: Naivete helps generate low liquid buffers and frequent credit-card borrowing, because the current self expects future selves to be more patient than they actually are.

4.5 Method of Simulated Moments

Let $\theta = (\beta, \delta, \rho)$ be the second-stage preference vector and χ the first-stage nuisance vector. Let $\bar{m}_{J_m}$ be the empirical moments and $m_{J_s}(\theta, \chi)$ the simulated moments. The simulated moment error is

$$g_{J_s}(\theta, \chi) = m_{J_s}(\theta, \chi) - \bar{m}_{J_m}. \quad (8)$$

The loss function is

$$q(\theta, \chi) = g_{J_s}(\theta, \chi)' W g_{J_s}(\theta, \chi), \quad (9)$$

and the estimator is

$$\hat{\theta} = \arg \min_{\theta} q(\theta, \hat{\chi}). \quad (10)$$

Interpretation: The model is estimated by making simulated balance sheets resemble observed SCF balance sheets, not by directly fitting consumption flows.

4.6 Standard error corrections

The second-stage variance includes corrections for first-stage estimation and simulation error:

$$\Omega_{\theta} = (G'_{\theta} W G_{\theta})^{-1} G'_{\theta} W (\Omega_g + \Omega_g^s + G_{\chi} \Omega_{\chi} G'_{\chi}) W G_{\theta} (G'_{\theta} W G_{\theta})^{-1}. \quad (11)$$

Interpretation: This matters empirically: Table 3 shows that correcting for first-stage uncertainty materially raises the standard error on $\hat{\beta}$.

5 Identification and empirical design

5.1 What identifies β , δ , and ρ

- Credit-card borrowing moments identify short-run impatience, because high-interest borrowing is hard to reconcile with a high short-run discount factor.
- Wealth moments identify long-run patience, because households accumulate substantial illiquid wealth over the lifecycle.
- Relative risk aversion affects precautionary saving, the value of liquidity, and the trade-off between consumption smoothing and borrowing.

Interpretation: The main identifying tension is that the household must be sufficiently impatient to borrow on expensive credit cards but sufficiently patient to build illiquid wealth.

5.2 Boundary analysis

The paper decomposes the MSM objective into contributions from credit-card moments and wealth moments:

$$q(\theta, \hat{\chi}) = q_{cc}(\theta, \hat{\chi}) + q_{wealth}(\theta, \hat{\chi}). \quad (12)$$

For different values of ρ , the paper plots regions in (β, δ) space that fit credit-card moments and regions that fit wealth moments.

Interpretation: This visualization makes the structural identification transparent. The model needs a point where the red credit-card region and the blue wealth region overlap or nearly overlap.

5.3 Limitations and interpretation

- The model abstracts from some forced-saving mechanisms, such as mortgage amortization and employer retirement defaults.
- Illiquid assets are modeled in reduced form and do not separately distinguish housing, pensions, and vehicles.
- Both benchmark present-bias and exponential models are formally rejected by overidentification tests because the moments are precise.
- The estimates depend on calibrated interest-rate wedges between credit-card debt and illiquid returns.

Interpretation: The results should be read as evidence that present bias helps explain a major balance-sheet pattern, not as a claim that this parsimonious lifecycle model exactly matches every moment.

6 Tables and figures

6.1 Table 1: Second-stage moments and Table 2: First-stage estimation results

Read:

- Table 1 reports the sixteen empirical moments used in second-stage estimation.
- The share revolving credit card debt is high at all working ages: 0.640 for ages 21-30 and 0.503 for ages 51-60.
- Average carried credit-card debt is smaller than income but persistent across age groups, around 0.096 to 0.111 of income.
- Households with credit card debt still hold wealth that rises sharply with age, from 1.222 times income at ages 21-30 to 4.650 at ages 51-60.
- Table 2 reports first-stage demographic, credit-limit, return, and income-process estimates.
- The benchmark real credit-card interest rate is 1.1059 annually, while the real return on positive liquid assets is 1.0203 and the illiquid asset consumption flow is 0.05.

Economic message: The first table states the puzzle; the second table fixes the institutional environment. The later preference estimates must explain high credit-card borrowing and large illiquid wealth without changing these first-stage primitives.

Name	$\bar{m}_{Jm}$	$se(\bar{m}_{Jm})$
% Visa:		
21-30	0.640	0.021
31-40	0.629	0.026
41-50	0.588	0.029
51-60	0.503	0.037
mean Visa:		
21-30	0.111	0.012
31-40	0.096	0.014
41-50	0.110	0.017
51-60	0.104	0.020
wealth debt:		
21-30	1.222	0.117
31-40	1.868	0.167
41-50	3.377	0.227
51-60	4.650	0.340
wealth no debt:		
21-30	1.659	0.129
31-40	2.800	0.154
41-50	4.613	0.252
51-60	8.071	0.393

(a) Table 1: Second-stage moments

Demographics Number of children $k_t = \phi_0 \exp(\phi_1 \text{age} - \phi_2 \frac{\text{age}^2}{100}) + \epsilon$			Liquid assets Credit limit λ_t $\lambda_t = \phi_0 + \phi_1 \text{age} + \phi_2 \frac{\text{age}^2}{100} + \epsilon$		
ϕ_0 0.003 (4.21E-06)	ϕ_1 0.358 (6.61E-05)	ϕ_2 0.508 (8.69E-05)	ϕ_0 0.167 (6.61E-02)	ϕ_1 -0.002 (3.18E-03)	ϕ_2 0.014 (3.22E-03)
Number of dependent adults $a_t = \phi_0 \exp(\phi_1 \text{age} - \phi_2 \frac{\text{age}^2}{100}) + \epsilon$			Real return on positive liquid assets R 1.0203 -		
ϕ_0 4.59E-06 (1.41E-08)	ϕ_1 0.452 (1.24E-04)	ϕ_2 0.438 (1.23E-04)	Real credit card interest rate R^{cc} 1.1059 -		
Illiquid assets Consumption flow as a fraction of assets γ 0.05 -			Real income from wages and transfers Income process $y_t = \ln(Y_t) = \phi_0 + \phi_1 \text{age} + \phi_2 \frac{\text{age}^2}{100} + \phi_3 \frac{\text{age}^3}{10000} + \phi_4 Nheads + \phi_5 Nchildren + \phi_6 Ndep.adults + \zeta_t$ $\zeta_t = \eta_t + \nu_t = \psi \eta_{t-1} + \epsilon_t + \nu_t$		
ϕ_0 7.563 (0.256)	ϕ_1 0.135 (0.017)	ϕ_2 -0.222 (0.037)	ϕ_3 0.106 (0.024)	ϕ_4 0.319 (0.023)	ϕ_5 0.013 (0.008)
			ϕ_6 0.237 (0.012)	ψ 0.840 (0.047)	σ_ϵ^2 0.057 (0.005)
					σ_ν^2 0.045 (0.004)

(b) Table 2: First-stage estimation results

6.2 Table 3: Benchmark estimates and Figure 1: q on β

Read:

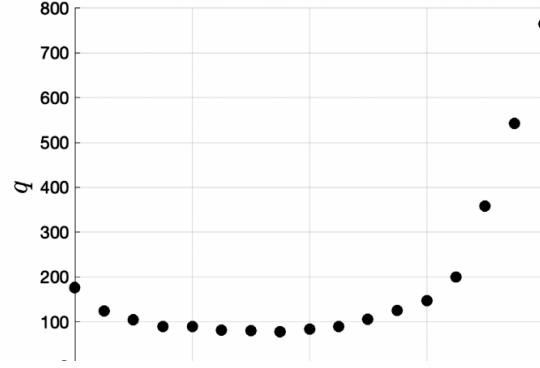
- Table 3 compares the benchmark present-biased model with the constrained exponential model.
- Present bias estimates $\hat{\beta} = 0.5305$, $\hat{\delta} = 0.9891$, and $\hat{\rho} = 1.9355$.
- The exponential model fixes $\beta = 1$ and estimates $\hat{\delta} = 0.9601$, but it underpredicts both the share and level of credit-card borrowing.
- The present-biased model's goodness-of-fit statistic is much lower than the exponential model's, though both are rejected in overidentification tests.
- Figure 1 shows that model fit worsens sharply as β approaches one, even when δ and ρ are reestimated.

Economic message: The key message is not merely that $\hat{\beta} < 1$. It is that the full balance-sheet pattern is hard to fit with exponential discounting at market interest rates.

Table 3
Benchmark estimates

	(1) Present Biased	(2) Exponential	(3) Data
Parameter estimates			
$\hat{\beta}$	0.5305	1	-
s.e. (i)	(0.1143)	-	-
s.e. (ii)	(0.1128)	-	-
s.e. (iii)	(0.0536)	-	-
s.e. (iv)	(0.0503)	-	-
$\hat{\delta}$	0.9891	0.9601	-
s.e. (i)	(0.0051)	(0.0053)	-
s.e. (ii)	(0.0051)	(0.0053)	-
s.e. (iii)	(0.0022)	(0.0022)	-
s.e. (iv)	(0.0020)	(0.0020)	-
$\hat{\rho}$	1.9355	1.4663	-
s.e. (i)	(0.4350)	(0.2256)	-
s.e. (ii)	(0.4280)	(0.2228)	-
s.e. (iii)	(0.2188)	(0.1142)	-
s.e. (iv)	(0.2045)	(0.1085)	-
Second-stage moments			
% Visa 21-30	0.605	0.309	0.640
% Visa 31-40	0.585	0.287	0.629
% Visa 41-50	0.523	0.299	0.588
% Visa 51-60	0.475	0.257	0.503
mean Visa 21-30	0.103	0.044	0.111
mean Visa 31-40	0.117	0.051	0.096
mean Visa 41-50	0.124	0.059	0.110
mean Visa 51-60	0.116	0.034	0.104
wealth 21-30 debt	0.913	0.880	1.222
wealth 31-40 debt	1.412	0.999	1.868
wealth 41-50 debt	2.640	1.941	3.377
wealth 51-60 debt	4.723	3.811	4.650
wealth 21-30 no debt	2.324	2.017	1.659
wealth 31-40 no debt	3.248	2.850	2.800
wealth 41-50 no debt	4.633	3.967	4.613
wealth 51-60 no debt	7.475	5.358	8.071
Goodness-of-fit			
$q(\hat{\theta}, \hat{\gamma})$	77.15	759.57	-
$\zeta(\hat{\theta}, \hat{\gamma})$	96.77	300.78	-

(a) Table 3: Benchmark estimates



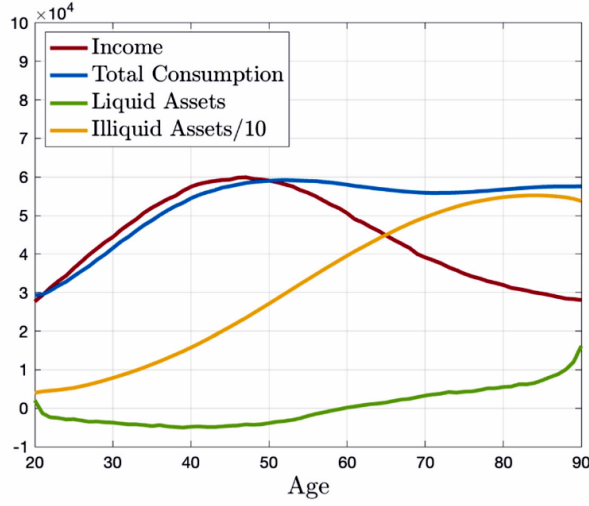
(b) Figure 1: q on β allowing δ and ρ to vary

6.3 Figure 2: Average lifecycle profile and Figure 3: Boundary analysis

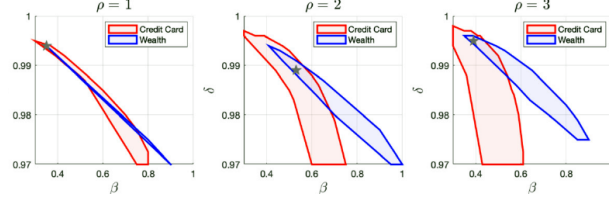
Read:

- Figure 2 plots model-implied lifecycle profiles for income, total consumption, liquid assets, and illiquid assets.
- Income peaks around middle age and then declines, while total consumption remains relatively smooth.
- Illiquid assets accumulate through working life; liquid assets are low or negative during many ages, reflecting credit-card borrowing.
- Figure 3 decomposes identification into credit-card and wealth moment regions.
- The red boundary is the region that fits credit-card moments; the blue boundary is the region that fits wealth moments.
- As ρ changes, the boundaries shift; the estimated $\rho \approx 1.9$ balances separation of the boundaries against the size of the feasible regions.

Economic message: Figure 2 explains the mechanism, while Figure 3 explains the identification. The household can look impatient in liquid assets and patient in illiquid assets because different parts of the balance sheet load on different parts of the discount function.



(a) Figure 2: Average lifecycle profile for present-biased estimate



(b) Figure 3: Boundary analysis

6.4 Table 4: Sources of identification

Read:

- Table 4 reestimates the model after dropping different moment blocks.
- Dropping mean credit-card debt, wealth conditional on debt, or wealth conditional on no debt shifts the estimates, but most specifications still imply $\beta < 1$.
- Using only wealth moments gives $\hat{\beta} = 0.7612$ and $\hat{\delta} = 0.9798$, which fits wealth relatively well but misses the borrowing facts.
- Using only the two credit-card moments gives $\hat{\beta} = 0.4827$ and $\hat{\delta} = 0.9893$, but the model no longer matches wealth accumulation.
- The shaded blocks show which empirical moments were not used in each column; comparing shaded and unshaded cells reveals which moment group drives each parameter.

Economic message: No single moment block carries the entire result. Credit-card moments and wealth moments pull the parameters in different directions, and the benchmark estimate is the compromise that best fits both.

Table 4
Sources of identification

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Benchmark	$\bar{meanVisa}$ $wealth debt$ $wealth no\ debt$	$\%Visa$ $wealth debt$ $wealth no\ debt$	$\bar{meanVisa}$ $wealth no\ debt$	$\bar{meanVisa}$ $wealth debt$ $wealth no\ debt$	$\bar{wealth debt}$ $wealth no\ debt$	$\bar{meanVisa}$ $wealth no\ debt$	Data
Parameter estimates								
$\hat{\beta}$	0.5305	0.6593	0.5309	0.2723	0.4088	0.7612	0.4827	-
s.e. ($\hat{\beta}$)	(0.1143)	(0.0760)	(0.0965)	(0.0602)	(0.0928)	(0.0871)	(0.1638)	-
$\hat{\delta}$	0.9891	0.9844	0.9886	0.9975	0.9925	0.9798	0.9893	-
s.e. ($\hat{\delta}$)	(0.0051)	(0.0039)	(0.0042)	(0.0016)	(0.0031)	(0.0053)	(0.0036)	-
$\hat{\rho}$	1.9355	1.9548	1.7870	3.9063	1.1567	1.8133	2.4276	-
s.e. ($\hat{\rho}$)	(0.4350)	(0.3582)	(0.6178)	(0.1327)	(0.1090)	(0.4516)	(1.5034)	-
Second-stage moments								
$\%Visa\ 21-30$	0.605	0.504	0.612	0.642	0.611	0.439	0.629	0.640
$\%Visa\ 31-40$	0.585	0.498	0.595	0.606	0.587	0.435	0.609	0.629
$\%Visa\ 41-50$	0.523	0.479	0.540	0.524	0.527	0.442	0.564	0.588
$\%Visa\ 51-60$	0.475	0.436	0.497	0.437	0.481	0.414	0.497	0.503
$mean\ Visa\ 21-30$	0.103	0.082	0.106	0.113	0.109	0.068	0.109	0.111
$mean\ Visa\ 31-40$	0.117	0.094	0.120	0.121	0.124	0.079	0.124	0.096
$mean\ Visa\ 41-50$	0.124	0.107	0.130	0.121	0.133	0.095	0.135	0.110
$mean\ Visa\ 51-60$	0.116	0.090	0.125	0.103	0.148	0.077	0.117	0.104
$wealth\ 21-30\ debt$	0.913	0.951	0.922	0.648	1.087	0.994	0.774	1.222
$wealth\ 31-40\ debt$	1.412	1.559	1.412	1.017	1.619	1.634	1.140	1.868
$wealth\ 41-50\ debt$	2.640	2.892	2.588	2.207	2.835	3.022	2.170	3.377
$wealth\ 51-60\ debt$	4.723	5.213	4.553	4.662	4.776	5.301	4.003	4.650
$wealth\ 21-30\ no\ debt$	2.324	2.256	2.348	2.069	2.515	2.210	2.188	1.659
$wealth\ 31-40\ no\ debt$	3.248	3.186	3.260	2.786	3.590	3.173	2.891	2.800
$wealth\ 41-50\ no\ debt$	4.633	4.689	4.628	4.288	4.937	4.649	4.128	4.613
$wealth\ 51-60\ no\ debt$	7.475	7.246	7.377	7.638	7.672	7.014	6.528	8.071
Goodness-of-fit								
$q(\hat{\theta}, \hat{\gamma})$	77.15	53.22	72.85	25.61	29.04	43.18	8.17	-

This table reports estimates of the discount function under alternative moment sets. Columns 2 through 7 vary the set of empirical moments used for estimation by dropping the shaded moments, maintaining all other benchmark assumptions. In Columns 2, 3, 4, and 5, we remove each of the four moment sets individually. In Column 6, we estimate using two conditional wealth moments, $wealth|debt$ and $wealth|no\ debt$. In Column 7, we estimate using two credit card borrowing moments, $\%Visa$ and $meanVisa$.

Table 4: Sources of identification

7 Short summary

- The paper estimates time preferences from lifecycle balance-sheet choices rather than experimental sooner-later choices.
- The key empirical moments combine credit-card borrowing and wealth accumulation across four age groups.
- The benchmark sample is households with a high-school degree and no college degree.
- The model has liquid assets, illiquid assets, credit cards, stochastic income, dependents, Social Security, mortality, and bequests.
- The benchmark present-biased estimate is $\hat{\beta} = 0.5305$, $\hat{\delta} = 0.9891$, and $\hat{\rho} = 1.9355$.
- Imposing exponential discounting produces a worse fit because the model cannot match frequent credit-card borrowing.
- The estimated short-run discounting rationalizes high-interest borrowing, while the high long-run discount factor rationalizes illiquid wealth accumulation.
- Boundary analysis shows how credit-card and wealth moments identify different parts of parameter space.
- Heterogeneity and sophistication extensions do not eliminate the need for $\beta < 1$ in the benchmark interest-rate environment.

8 Conclusion

8.1 Core conclusion

The paper estimates a discount function from large-stakes lifecycle financial choices and finds strong evidence of present bias. The household balance sheet contains two facts that are hard to reconcile with a single exponential discount factor: many households borrow on high-interest credit cards, yet they also accumulate substantial illiquid wealth. A present-biased model with low short-run patience and high long-run patience fits this tension much better than a conventional exponential model.

8.2 Economic mechanism

The mechanism is balance-sheet segmentation combined with present-focused preferences. When households experience high liquidity, they often place resources into illiquid assets that earn a premium or provide consumption services. Later, because current selves overconsume relative to previous plans, liquid balances can fall and households borrow on credit cards even while they retain illiquid wealth. This generates the coexistence of debt and wealth.

8.3 Incremental contribution relative to prior work

Relative to experiments on intertemporal choice, the paper estimates time preferences from consequential field choices. Relative to buffer-stock savings models, it adds present-biased preferences and a realistic split between liquid and illiquid assets. Relative to earlier present-bias models, it uses a formal two-stage MSM estimator and a transparent identification diagnostic. The key contribution is to show that present bias is not just a behavioral story; it can be quantitatively disciplined by lifecycle balance-sheet data.

8.4 What the paper teaches about structural estimation

The paper shows that a rich structural model can be made interpretable if the author decomposes the objective function into moment groups. The boundary analysis is valuable because it lets the reader see why the model chooses a low β , a high δ , and an intermediate ρ instead of treating the estimates as a black box.

8.5 Policy interpretation

The estimates have implications for credit-card regulation, retirement account liquidity, and default options. More liquidity in retirement accounts could reduce expensive credit-card borrowing, but it may also undermine commitment value for present-biased households. The welfare effect therefore depends on whether illiquidity is mainly a friction or a self-control device.

8.6 Limitations and open questions

The model treats illiquid wealth in reduced form and cannot fully distinguish housing equity, retirement plans, vehicles, and other assets. It also does not model all institutional mechanisms that force saving, such as mortgage amortization, payroll deductions, and employer defaults. The formal overidentification tests reject both benchmark models, suggesting that the present-biased model captures an important mechanism but not all household balance-sheet behavior.

8.7 Takeaway

The paper's central lesson is that household patience is horizon dependent. A single exponential discount factor cannot easily explain both expensive borrowing and long-run wealth accumulation. A discount function with present bias gives a compact quantitative account of why households can look impatient in liquid accounts and patient in illiquid accounts at the same time.